Joint-Cancellation Full-Trace Transfer: An Exact First-Alias Criterion and Sharp Replacement Obstructions

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Abstract

The first-alias route has an exact local packet and a validated isolated model, but no theorem identifies that model with the actual noisy trace. We formulate the missing transfer criterion without making the identification. On a fixed-phase clock, a weighted full-trace prefix containing the first alias but no second alias splits exactly into an off-alias background and $|e_{2k}|/(2H_k)$, where $H_k = kR^{-2k}$. For the observable five-slot packet $e = \mathcal{B}+\mathcal{S}+\mathcal{R}+\mathcal{P}-\mathcal{A}$, actual minus model replacement gives the exact identity

$$e_k^{\mathrm{a}} = \widehat{e}_k + \{\Delta\mathcal{B}_k + \Delta\mathcal{S}_k + \Delta\mathcal{R}_k + \Delta\mathcal{P}_k - \Delta\mathcal{A}_k\}.$$

Thus a closing model transfers exactly when the signed joint defect is $o(H_k)$. We retain all $4k$ two-channel Duhamel terms and prove a sharp grouped signed interval certificate. Separate absolute majorants are sufficient but not necessary: equal defects of size $\mathcal{A}_k$ may cancel or reinforce. Applied to the failed RH-329 model, actual closure requires an $\mathcal{A}_k$-sized correction tuned to H_k precision; an explicit synthetic repair shows why isolated failure does not imply actual full-trace divergence. The physical identification, signed Duhamel enclosures, far remainder, and off-alias background remain open, so the criterion is not activated and no Riemann-hypothesis conclusion follows.

1 One-alias full-trace extraction

Work along a fixed-phase moving-order subsequence

$$k = k_\sigma = \frac{\log(1/\sigma)}{2\log\lambda} + O(1), \qquad \eta_\sigma \longrightarrow \eta, \qquad H_k = kR^{-2k}. \tag{1}$$

Let $q_{\sigma,k,n}$ denote the typed coefficient error at order n for the actual full-trace comparison, and suppose its critical coefficient at $n = 2k$ is the signed packet e_k^{a} defined below. Choose an integer cut h_σ with

$$2k < h_\sigma \le 4k. \tag{2}$$

The weighted prefix and its off-alias part are

$$\mathcal{E}_\sigma^{(h)}(R) = \sum_{2\le n<h_\sigma} \frac{|q_{\sigma,k,n}|R^n}{n}, \tag{3}$$

$$\mathcal{E}_{\sigma,\mathrm{off}}^{(h)}(R) = \sum_{\substack{2\le n<h_\sigma\\ n\ne 2k}} \frac{|q_{\sigma,k,n}|R^n}{n}. \tag{4}$$

These quantities are nonnegative.

Theorem 1.1 (Exact critical extraction)**.** *On the window* (2)*,*

$$\boxed{\mathcal{E}_\sigma^{(h)}(R) = \mathcal{E}_{\sigma,\mathrm{off}}^{(h)}(R) + \frac{|e_k^{\mathrm{a}}|}{2H_k}.} \tag{5}$$

Consequently,

$$\mathcal{E}_\sigma^{(h)}(R) \longrightarrow 0 \quad\Longleftrightarrow\quad \mathcal{E}_{\sigma,\mathrm{off}}^{(h)}(R) \longrightarrow 0 \quad \textit{and} \quad e_k^{\mathrm{a}} = o(H_k). \tag{6}$$

Proof. Separate the $n = 2k$ summand in (3). Since $H_k = kR^{-2k}$,

$$\frac{|e_k^{\mathrm{a}}|R^{2k}}{2k} = \frac{|e_k^{\mathrm{a}}|}{2H_k}.$$

The equivalence follows because both terms on the right of (5) are nonnegative. □

This theorem isolates the full-trace constituent controlled by RH-330. It does not prove that the off-alias term vanishes. Nor does it close the separate head/counterloop budget required by determinant gluing.

2 Observable signed transfer

At the critical order, retain the actual localized slots from RH-327 and the parity/alias signs from RH-326:

$$e_k^{\mathrm{a}} = \mathcal{B}_k^{\mathrm{a}} + \mathcal{S}_k^{\mathrm{a}} + \mathcal{R}_k^{\mathrm{a}} + \mathcal{P}_k^{\mathrm{a}} - \mathcal{A}_k^{\mathrm{a}}. \tag{7}$$

Let a frozen comparison model have the same data types,

$$\widehat{e}_k = \widehat{\mathcal{B}}_k + \widehat{\mathcal{S}}_k + \widehat{\mathcal{R}}_k + \widehat{\mathcal{P}}_k - \widehat{\mathcal{A}}_k. \tag{8}$$

Define actual-minus-model defects

$$\Delta\mathcal{Y}_k = \mathcal{Y}_k^{\mathrm{a}} - \widehat{\mathcal{Y}}_k, \qquad \mathcal{Y} \in \{\mathcal{B}, \mathcal{S}, \mathcal{R}, \mathcal{P}, \mathcal{A}\}, \tag{9}$$

and their signed joint aggregate

$$\Theta_k = \Delta\mathcal{B}_k + \Delta\mathcal{S}_k + \Delta\mathcal{R}_k + \Delta\mathcal{P}_k - \Delta\mathcal{A}_k. \tag{10}$$

Theorem 2.1 (Sharp signed transfer criterion)**.** *For every typed actual/model pair on the same clock,*

$$\boxed{e_k^{\mathrm{a}} = \widehat{e}_k + \Theta_k.} \tag{11}$$

In particular:

$$e_k^{\mathrm{a}} = o(H_k) \quad\Longleftrightarrow\quad \Theta_k = -\widehat{e}_k + o(H_k); \tag{12}$$

$$\widehat{e}_k = o(H_k) \quad\Longrightarrow\quad \{e_k^{\mathrm{a}} = o(H_k) \Longleftrightarrow \Theta_k = o(H_k)\}. \tag{13}$$

If $\Theta_k/H_k \to 0$, then $\widehat{e}_k/H_k$ and e_k^{a}/H_k have the same finite or extended-real limit. Combined with theorem 1.1, actual weighted-prefix closure follows from

$$\mathcal{E}_{\sigma,\mathrm{off}}^{(h)}(R) \to 0, \qquad \widehat{e}_k = o(H_k), \qquad \Theta_k = o(H_k). \tag{14}$$

Proof. Subtract (8) from (7) to obtain (11). Dividing by H_k proves (12), (13), and the limit statement. The last assertion is (6). □

The minimal condition (12) permits cancellation between every slot. A stronger modular condition, useful for an auditable physical theorem, is

$$\Theta_k^{\mathrm{crit}} := \Delta\mathcal{B}_k + \Delta\mathcal{S}_k + \Delta\mathcal{P}_k - \Delta\mathcal{A}_k = o(H_k), \tag{15}$$

$$\Delta\mathcal{R}_k = o(H_k). \tag{16}$$

It implies $\Theta_k = o(H_k)$ but is not logically necessary. The point of (16) is to control the already aggregated signed far trace rather than let an unknown far term repair a critical mismatch accidentally. It does not require separate absolute estimates for every microscopic far piece.

3 Exchange gauge and the observable shell

RH-328 writes a proposed shell representation as

$$\widehat{\mathcal{S}}_k = \widehat{\mathcal{X}}_k + \widehat{\mathcal{O}}_k, \qquad \widehat{\mathcal{X}}_k = L_k(\widehat{c}_k^{2k} - c_{0,k}^{2k}), \tag{17}$$

where $\widehat{\mathcal{O}}_k$ is the observation error. This split has a basic gauge freedom.

Proposition 3.1 (Exchange–observation gauge)**.** *For every real t_k, the transformation*

$$\widehat{\mathcal{X}}_k \longmapsto \widehat{\mathcal{X}}_k + t_k, \qquad \widehat{\mathcal{O}}_k \longmapsto \widehat{\mathcal{O}}_k - t_k \tag{18}$$

leaves $\widehat{\mathcal{S}}_k$, $\widehat{e}_k$, Θ_k, and (11) *unchanged. Therefore, unless a physical identification map freezes* (17) *before evaluation, the intrinsic shell replacement is*

$$\Delta\mathcal{S}_k = \mathcal{S}_k^{\mathrm{a}} - (\widehat{\mathcal{X}}_k + \widehat{\mathcal{O}}_k), \tag{19}$$

not separate claims about $\Delta\mathcal{X}_k$ and $\Delta\mathcal{O}_k$.

Proof. The two changes in (18) sum to zero. Every listed quantity depends on them only through $\widehat{\mathcal{S}}_k = \widehat{\mathcal{X}}_k + \widehat{\mathcal{O}}_k$. □

This prevents an after-the-fact redistribution between *exchange* and *observation error* from manufacturing a small component certificate.

4 All-term Duhamel transfer and grouped enclosures

Let $\chi \in \{-,+\}$ denote the two critical channels. In channel χ, let $A_{\chi,j}$ be an actual leg, $G_{\chi,j}$ its frozen comparison leg, and let the product length be $m_{\chi,k} = 2k$. For a bounded cyclic trace observation $\mathfrak{t}_\chi$, the exact hybrid terms are

$$d_{\chi,j} = \mathfrak{t}_\chi(A_{\chi,2k}\cdots A_{\chi,j+1}(A_{\chi,j} - G_{\chi,j})G_{\chi,j-1}\cdots G_{\chi,1}), \qquad 1 \le j \le 2k. \tag{20}$$

The product difference is $\sum_{j=1}^{2k} d_{\chi,j}$. If $|\mathfrak{t}_\chi(T)| \le T_\chi\|T\|$ and $\delta_{\chi,j} = \|A_{\chi,j} - G_{\chi,j}\|$, then

$$|d_{\chi,j}| \le W_{\chi,j}\delta_{\chi,j}, \tag{21}$$

$$W_{\chi,j} = T_\chi \prod_{\ell=j+1}^{2k} \|A_{\chi,\ell}\| \prod_{\ell=1}^{j-1} \|G_{\chi,\ell}\|. \tag{22}$$

Thus all $4k$ weights are retained. Fixed orientation signs and structural replacement terms are absorbed into a finite signed list $\{\xi_{i,k}\}_{i\in I_k}$ whose sum is Θ_k.

Absolute summation of (21) is a safe fallback, but it erases the cancellation sought by the first-alias route. The following certificate keeps it.

Theorem 4.1 (Sharp grouped signed enclosure)**.** *Partition I_k into groups G and suppose certified centers and radii obey*

$$\left|\sum_{i\in G} \xi_{i,k} - c_{G,k}\right| \le r_{G,k}, \qquad r_{G,k} \ge 0. \tag{23}$$

Put $C_k = \sum_G c_{G,k}$ and $\rho_k = \sum_G r_{G,k}$. Then

$$\Theta_k \in [C_k - \rho_k, C_k + \rho_k] \tag{24}$$

and

$$e_k^{\mathrm{a}} \in [\widehat{e}_k + C_k - \rho_k, \widehat{e}_k + C_k + \rho_k]. \tag{25}$$

The best and worst absolute residuals allowed by these data are exactly

$$\mathrm{best}_k = \max\{|\widehat{e}_k + C_k| - \rho_k, 0\}, \tag{26}$$
$$\mathrm{worst}_k = |\widehat{e}_k + C_k| + \rho_k. \tag{27}$$

Hence $|C_k| + \rho_k = o(H_k)$ *robustly certifies* $\Theta_k = o(H_k)$, *while* $|\widehat{e}_k + C_k| + \rho_k = o(H_k)$ *directly certifies actual closure.*

Proof. Sum (23) and apply the triangle inequality to obtain (24); then use (11). The distance from zero to a centered real interval gives (26), and its farthest endpoint gives (27). Without further cross-group information both endpoints can be attained, so the interval is sharp from the stated data. □

Taking singleton groups, zero centers, and radii $W_{\chi,j}\delta_{\chi,j}$ recovers the usual separate absolute Duhamel majorant. The theorem does not assert that the actual signed group enclosures exist; deriving them is an open physical problem.

5 Sharp scale obstructions

The $o(H_k)$ scale in theorem 2.1 cannot be weakened by a generic boundedness statement.

Proposition 5.1 (Four transfer obstructions)**.** *Let* $H_k > 0$.

1. *If* $\widehat{e}_k = 0$ *and* $\Theta_k = H_k$, *then* $e_k^{\mathrm{a}}/H_k = 1$. *Thus* $\Theta_k = O(H_k)$ *is insufficient for little-o closure.*
2. *Suppose* $\mathcal{A}_k/H_k \to \infty$ *at the RH-329 exponential rate. Taking* $\Delta\mathcal{B}_k = \mathcal{A}_k/k$ *and all other defects zero gives* $\Theta_k/\mathcal{A}_k = 1/k \to 0$ *but* $\Theta_k/H_k \to \infty$. *Hence* $o(\mathcal{A}_k)$ *fidelity is insufficient to transfer a closing model.*
3. *The two defect vectors*

$$(\Delta\mathcal{B}_k, \Delta\mathcal{S}_k) = (\mathcal{A}_k, -\mathcal{A}_k) \quad \textit{and} \quad (\mathcal{A}_k, \mathcal{A}_k)$$

   have identical componentwise absolute bounds. Their joint defects are, respectively, 0 *and* $2\mathcal{A}_k$. *Separate unsigned envelopes alone decide neither cancellation nor failure.*
4. *If a certificate omits the far slot, setting only* $\Delta\mathcal{R}_k = H_k$ *changes a closing model into a nonclosing actual packet. Every signed slot must be retained.*

Proof. The first, third, and fourth statements follow by substitution into (10). For the second, RH-329 gives $\mathcal{A}_k/H_k = (a_k/k)g^k$ with $g = (\beta R)^2 > 1$ and a_k/k tending to a positive constant. Therefore $\mathcal{A}_k/(kH_k) \to \infty$. □

The first defect vector in item 3 has separate absolute budget $2\mathcal{A}_k \gg H_k$ but transfers perfectly. This proves that separate absolute-majorant smallness is sufficient, not necessary.

6 The RH-329 repair and no-go transfer laws

For the exact model-defining data of RH-329, write

$$\varrho = 1 - C_* C_M \in (0,1). \tag{28}$$

Its proved asymptotics are

$$\frac{\widehat{e}_k}{\mathcal{A}_k} \longrightarrow -\varrho, \qquad \frac{\mathcal{A}_k}{H_k} \longrightarrow \infty. \tag{29}$$

Corollary 6.1 (Exact repair law and conditional no-go transfer)**.** *An actual packet compared with RH-329 closes at the target scale if and only if*

$$\Theta_k = -\widehat{e}_k + o(H_k). \tag{30}$$

In particular,

$$\frac{\Theta_k}{\mathcal{A}_k} \longrightarrow \varrho, \qquad \frac{\Theta_k + \widehat{e}_k}{\mathcal{A}_k} = o\left(\frac{H_k}{\mathcal{A}_k}\right) \tag{31}$$

are necessary. Conversely, if $\Theta_k = o(\mathcal{A}_k)$, then

$$\frac{e^{\mathrm{a}}_k}{\mathcal{A}_k} \longrightarrow -\varrho, \qquad \frac{e^{\mathrm{a}}_k}{H_k} \longrightarrow -\infty. \tag{32}$$

Proof. Equation (30) is (12). Divide it first by $\mathcal{A}_k$ and then use (29) to obtain (31). If $\Theta_k = o(\mathcal{A}_k)$, divide (11) by $\mathcal{A}_k$; the negative nonzero limit then multiplies the divergent ratio $\mathcal{A}_k/H_k$. ☐

There is an exact synthetic repair:

$$\Delta\mathcal{B}_k = -\widehat{e}_k + \frac{H_k}{k}, \qquad \Delta\mathcal{S}_k = \Delta\mathcal{R}_k = \Delta\mathcal{P}_k = \Delta\mathcal{A}_k = 0. \tag{33}$$

It gives $e^{\mathrm{a}}_k = H_k/k = o(H_k)$. This is a scalar transfer ledger, not an operator construction. Its role is logical: the RH-329 isolated no-go does not imply actual failure unless one proves a replacement condition such as (32). Likewise, constructing the scalar repair does not show that the actual operator closes.

7 Exact reproduction ledger

The executable audit imports the exact RH-329 packet values and performs all sign, interval, repair, and counterexample decisions with rational arithmetic. Directed decimals are presentation only. For each reported order it records all $4k$ synthetic signed Duhamel terms before cancellation; the six rows contain 344 terms total.

k	$\widehat{e}_k/H_k$	$\Theta_k^{\mathrm{repair}}/\mathcal{A}_k$	$H_k/\mathcal{A}_k$	$2\mathcal{A}_k/H_k$
2	-2.32697	0.714993	0.252918	7.90769
8	-42.3139	0.793551	0.0186987	106.959
16	-1899.64	0.805885	0.000424217	4714.57
32	-3.97557×10^6	0.801015	2.01484×10^{-7}	9.92634×10^6

Every repair row satisfies $e^{\mathrm{a}}_k/H_k = 1/k$ exactly. Every balanced row has signed defect zero, while its unsigned budget is $2\mathcal{A}_k/H_k$. These are formula checks and sharp synthetic examples, not finite-order evidence for physical asymptotics.

8 Inactive hypotheses and RH-331 handoff

The repository does not provide a map identifying actual blocks with RH-329 blocks. It also lacks a physical exchange–observation split, signed two-channel Duhamel group enclosures, target-scale parity and alias replacement, a signed far-remainder theorem, and convergence of the off-alias weighted background. RH-329 is a graded family, not one operator realizing every order. Therefore neither (14) nor (32) is activated for the actual noisy trace.

RH-331 should review RH-322–RH-330 as a ten-layer chain, preserve the observable-shell gauge firewall, verify the individual and batch archives, and record the route coordinate

`first_alias_transfer_criterion_exact_actual_replacement_open`.

The all-order deterministic envelope and coefficient anchors remain inherited inputs; they are not re-proved here. The open off-alias and head/counterloop budgets mean determinant gluing is not activated.

All five program gates remain false/open. No canonical intrinsic dynamical determinant, oriented unitary completion, self-adjoint generator with an intrinsic $T \log T$ law, von Mangoldt prime-power trace, or completed-zeta divisor equality is proved. No Hilbert–Polya operator is constructed, no Riemann zero is identified, and the Riemann Hypothesis is not proved.